Earners AI and ML - by Aditya Jain sir - Lecture 09

Link to Telegram Community: https://t.me/EarnersAdityaJain

Recap

```
for i in range(1,10):
  print(i)
```

```
1
2
3
4
5
6
7
8
9
```

```
for i in range(1,10):

  if i == 4:
    break

  print(i)
```

```
1
2
3
```

```
for i in range(1,10):

  if i == 4:
    continue

  print(i)
```

```
1
2
3
5
6
7
8
9
```

```
#infinite loop
i = 1
while i<= 10:

  if i == 4:
    continue

  print(i)
  i = i+1
```

```
1
2
3
---------------------------------------------------------------------------
KeyboardInterrupt                         Traceback (most recent call last)
/tmp/ipykernel_609/4080418503.py in <cell line: 0>()
      3
      4   if i == 4:
----> 5     continue
      6
      7   print(i)

KeyboardInterrupt:
```

```
i = 1
while i<= 10:

  if i == 4:
    i = i+1
    continue

  print(i)
  i = i+1
```

```
1
2
3
5
6
7
8
9
10
```

```
i = 1
while i<= 10:

  if i == 4:
    i = 5
    continue

  print(i)
  i = i+1
```

```
1
2
3
5
6
7
8
9
10
```

Nested Loops

```
for i in range(5):
  print("i=", i)
```

```
i= 0
i= 1
i= 2
i= 3
i= 4
```

```
for i in range(5):
  for j in range(3):
    print("i=", i, "j=", j)
```

```
i= 0 j= 0
i= 0 j= 1
i= 0 j= 2
i= 1 j= 0
i= 1 j= 1
i= 1 j= 2
```

```
i= 2 j= 0
i= 2 j= 1
i= 2 j= 2
i= 3 j= 0
i= 3 j= 1
i= 3 j= 2
i= 4 j= 0
i= 4 j= 1
i= 4 j= 2
```

```
  for j in range(3):
    print("j=", j)
```

```
j= 0
j= 1
j= 2
```

```
* * * * *

* * * * *

* * * * *

* * * * *
```

```
for j in range(5):
  print("*", end = " ")
```

```
* * * * *
```

```
for row in range(4):

  for col in range(5):
    print("*", end = " ")

  print()
```

```
* * * * *
* * * * *
* * * * *
* * * * *
```

```
1 1 1 1 1

2 2 2 2 2

3 3 3 3 3

4 4 4 4 4

5 5 5 5 5
```

```
for i in range(5):
  print(1, end = " ")
```

```
1 1 1 1 1
```

```
for i in range(5):
  print(2, end = " ")
```

```
2 2 2 2 2
```

```
for i in range(5):
  print(3, end = " ")
```

```
3 3 3 3 3
```

```
for i in range(1,6):

  for j in range(5):
    print(i, end = " ")

  print()
```

```
1 1 1 1 1
2 2 2 2 2
3 3 3 3 3
4 4 4 4 4
5 5 5 5 5
```

```
1 2 3 4 5

1 2 3 4 5

1 2 3 4 5

1 2 3 4 5

1 2 3 4 5
```

```
for j in range(1,6):
  print(j, end = " ")
```

```
1 2 3 4 5
```

```
for i in range(5):

  for j in range(1,6):
    print(j, end = " ")

  print()
```

```
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
```

```
10 20 30 40
10 20 30 40
10 20 30 40
```

```
for i in range(1,5):
  print(i, end = " ")
```

```
1 2 3 4
```

```
for i in range(1,5):
  print(i*10, end = " ")
```

```
10 20 30 40
```

```
for j in range(3):

  for i in range(1,5):
    print(i*10, end = " ")

  print()
```

```
10 20 30 40
10 20 30 40
10 20 30 40
```

```
*
* *
* * *
* * * *
* * * * *
```

```
#Pattern:

row 1 --> 1
row 2 --> 2
row 3 --> 3
```

```
for row in range(1,6):

  for col in range(row):
    print("*", end = " ")

  print()
```

```
*
* *
* * *
* * * *
* * * * *
```

```
1
2 2
3 3 3
4 4 4 4
5 5 5 5 5
```

```
row1 --> 1 -1
row2 --> 2-2
row3 --> 3-3
```

```
for i in range(1,6):

  for j in range(i):
    print(i, end = " ")

  print()
```

```
1
2 2
3 3 3
4 4 4 4
5 5 5 5 5
```

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

```
row1 --> 1 to 1
row2 --> 1 to 2
row3 --> 1 to 3
```

```
for i in range(1,6):

  for j in range(1, i+1):
    print(j, end = " ")

  print()
```

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

```
1
2 3
4 5 6
7 8 9 10
```

```
n = 1

for row in range(1, 5):

  for col in range(row):
    print(n, end = " ")
    n = n + 1
```

```
  print()
```

```
1
2 3
4 5 6
7 8 9 10
```

Quiz

```
What is a nested loop?

A. A loop that runs forever

B. A loop inside another loop

C. Two loops written separately

D. A loop inside an if statement


Ans: B
```

```
How many total times will the * get printed?

for i in range(3):
    for j in range(4):
        print("*")

 Ans: 12
```

```
What is the output?
for i in range(2):
    for j in range(3):
        print(i, j)

A.

0 0
0 1
0 2
1 0
1 1
1 2

B.

0 1
2 3

C.

0
1
2

D. Error


Ans: A
```

Which loop controls the number of rows in most pattern-printing programs?

A. Inner loop

B. Outer loop

C. if statement

D. break

Ans: B

Which loop controls what gets printed in each row?

A. Outer loop

B. Inner loop

C. while

D. continue

Ans: B

What is the output?

```
for row in range(3):
    for col in range(2):
        print("*", end=" ")
    print()
```

A.

```
* *
* *
* *
```

B.

```
******
```

C.

```
*
*
*
```

D. Error

Ans: A

What does print() without arguments do inside a nested loop?

A. Stops the program

B. Moves to the next line

C. Skips a row

D. Ends the loop

Ans: B

What is the output?

```
for i in range(1,4):
    for j in range(i):
        print("*", end=" ")
    print()
```

```
A.

*
* *
* * *

B.

***
**
*

C.

*
**
***

D. Both A and C


Ans: A
```

```
What is printed?

for i in range(1,4):
    for j in range(1,4):
        print(i, end=" ")
    print()

A.

1 2 3
1 2 3
1 2 3

B.

1 1 1
2 2 2
3 3 3

C.

1
2
3

D. Error



Ans: B
```

```
for i in range(1,4):
    for j in range(1,4):
        print(i, end=" ")
    print()
```

```
1 1 1
2 2 2
3 3 3
```

```
for i in range(1,4):
    for j in range(1,4):
        print(j, end=" ")
    print()
```

```
1 2 3
1 2 3
1 2 3
```

```
What is the output?
for i in range(2):
    for j in range(2):
        print(j, end=" ")
    print()

A.

0 1
0 1

B.

1 2
1 2

C.

0
1

D. Error
```

```
Which pattern will this code print?
for row in range(4):
    for col in range(4):
        print("#", end=" ")
    print()

A.

#
##
###
####

B.

# # # #
# # # #
# # # #
# # # #

C.

####
####
####
####

D.

#
#
#
#
```

```
What is the output?
for i in range(1,3):
    for j in range(1,3):
        print(i + j, end=" ")
    print()

A.

2 3
3 4

B.

1 2
2 3
```